Analyzing E-Learning Platform purchases

Project Overview

This project focuses on analyzing purchase data from an E-Learning platform using MySQL. The dataset includes information about learners, courses, and their purchase transactions. By combining and analyzing these datasets, the project aims to understand user behavior, course performance, and revenue generation.

The database is designed with relational tables such as learners, courses, and purchases, connected using primary and foreign keys to maintain data integrity. Various SQL queries are used to extract meaningful insights from the data.

Project Objective

- To analyze learner purchasing behavior
- To identify top-performing and low-performing courses
- To calculate total revenue and category-wise revenue
- To track sales trends based on purchase dates
- To find the most active learners on the platform
- To understand course demand across different categories
- To generate insights that can help improve business decisions

Dataset Description

· **Learners Table**
Contains information about users such as learner ID, name, and country.
· **Courses Table**
Includes course details like course ID, course name, category, and unit price.
· **Purchases Table**
Stores transaction data including purchase ID, learner ID, course ID, quantity, and purchase date.

Problem Statement

- In today's digital learning environment, E-Learning platforms generate large volumes of data related to learners, courses, and purchase transactions. However, without proper analysis, this data does not provide meaningful insights for decision-making.
- The challenge is to analyze the available data to understand learner behavior, course performance, and revenue trends. This includes identifying which courses are most popular, which generate the highest revenue, how learners interact with different categories, and how sales vary over time.
- By using SQL queries on structured relational data, the goal is to transform raw data into actionable insights that can help improve course offerings, enhance user engagement, and support better business strategies.

Attributes (Tables & Columns Details)

LEARNERS TABLE

Attribute	Type	Description
Learner_id	INT (PK)	Unique identifier for each laerner
Full_name	VARCHAR	Name of the learner
country	VARCHAR	Country of the learner

COURSES TABLE

Attribute	Type	Description
Course_id	INT (PK)	Unique identifier for each course
Course_name	VARCHAR	Name of the course
category	VARCHAR	Category of the course
Unit_price	INT	Price of the course

PURCHASES TABLE

Attribute	Type	Description
Purchase_id	INT(PK)	Unique identifier for each purchase
Learner_id	INT (FK)	References learner who made the purchase
Course_id	INT (FK)	References purchased course
quantity	INT	Number of courses purchased
Purchase_date	DATETIME	Date and time of purchase

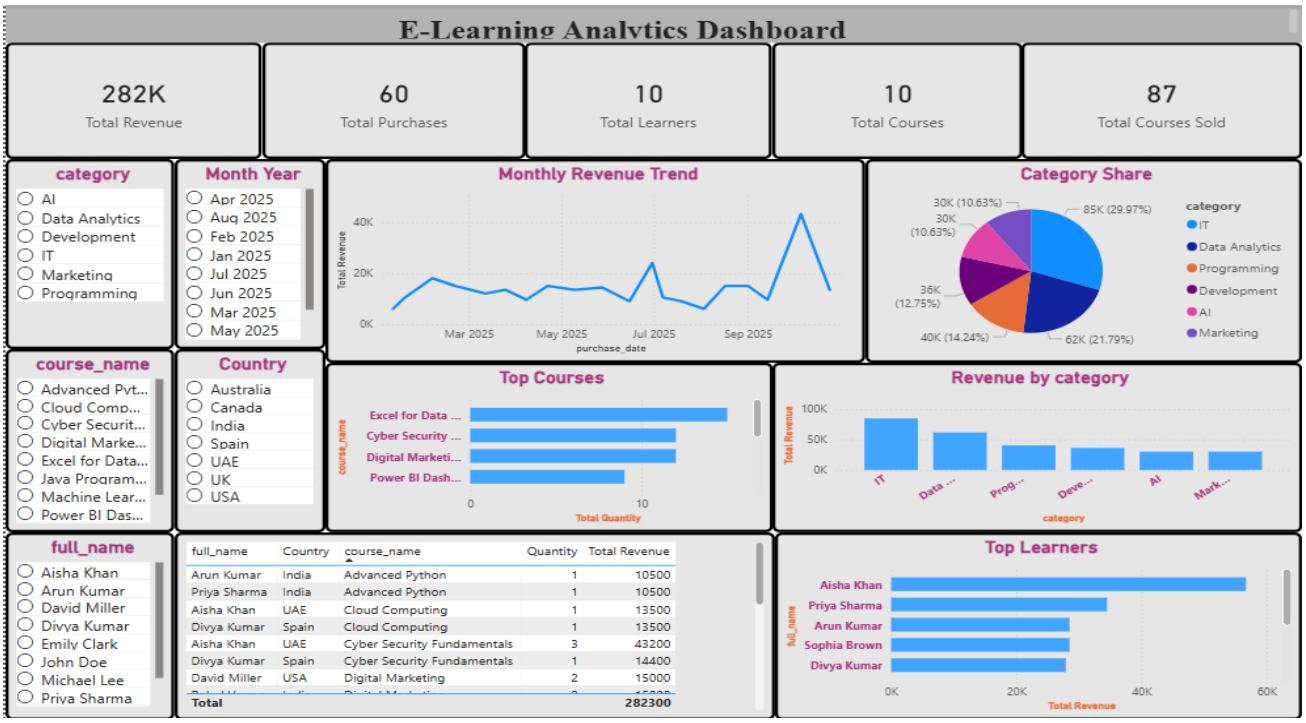
Insights (SQL Analysis)

- Aisha Khan is the highest revenue-generating learner ₹56700
- John Doe is the most active buyer (highest number of purchases)
- Cyber Security Fundamentals is the highest revenue-generating course ₹57600
- Excel for Data Analysis is the most purchased course (highest total quantity sold)
- SQL for Beginners is the lowest revenue-generating course
- IT category generates the highest overall revenue
- India has the highest number of learners and total purchases
- October records the highest monthly revenu
- No course is completely unsold (all courses have at least one purchase)
- Multiple learners purchased from more than one category

Visualization (Power BI)

Key Insights

- IT category contributes the highest revenue (~30%)
- Data Analytics is the second-highest revenue category
- Marketing category generates the lowest revenue share
- Revenue shows a steady increase with a peak towards the end of the year
- September–October period records the highest revenue spike
- Top learners contribute significantly higher revenue than others
- Course enrollments are unevenly distributed across categories
- With only 87 courses sold generating 282K revenue, the platform shows high revenue per sale
- The dashboard shows strong performance in technical domains (IT & Analytics)



Recommendtion

- Focus more on IT & Data Analytics courses
- Improve low-performing categories (Marketing, AI)
- Run campaigns during peak months (Sep–Oct)
- Offer course bundles to increase sales
- Reward top customers to boost retention
- Update or remove low-performing courses
- Optimize pricing strategy for better revenue

Conclusion

The platform shows strong demand for technical and analytics courses, with revenue driven by both high-volume low-cost courses and premium IT offerings, while also highlighting opportunities in underperforming categories and geographic expansion.”